SR-MPLS Training Lab

M2 SR-MPLS Transport

Estimated time: 60 min

Objective

Turn the IS-IS underlay into a Segment Routing MPLS transport layer. You will enable SPRING (source-packet-routing) on top of IS-IS, define the shared label block, give each node a unique node-SID, and confirm that every loopback is now reachable over a label-switched path instead of plain IP. By the end PE1 can build and trace an MPLS tunnel to PE2 using label 16002, which is the transport that L3VPN services in later modules ride on.

Before you start: M1 complete: IS-IS Level 2 is up on every node, all adjacencies read Up, and loopback-to-loopback IP reachability is proven, so the SPF tree SR will reuse is already correct.

Concept primer

Segment Routing runs as an extension of IS-IS, not a separate protocol. Junos calls it SPRING, configured under 'protocols isis source-packet-routing'. There is no LDP or RSVP signalling: IS-IS floods a few new sub-TLVs in the LSPs you already have, and every router computes the same label bindings from the same link-state database. The source router decides the whole path by pushing a stack of labels called segments, which is why it is 'source' packet routing. In this lab a single node-SID per destination is enough, so each tunnel is a one-label stack.

All routers share one label block called the SRGB (Segment Routing Global Block). This lab sets 'srgb start-label 16000 index-range 8000', so the block runs from label 16000 to 23999 and is identical on every node. Each router is then given a node-segment 'ipv4-index', and its absolute node-SID label is simply SRGB-start plus that index. PE1 uses index 1, so its label is 16001; PE2 uses index 2, so its label is 16002. Because the SRGB is global and consistent, label 16002 means 'the path to PE2' from the point of view of every router in the domain. The index must be globally unique: two routers sharing an index both claim the same label, which is the fault this module's break/fix explores.

Two moving parts complete the picture. 'explicit-null' tells upstream neighbors to send the explicit-null label (label 0) on the final hop instead of popping the label

entirely (penultimate-hop popping), which keeps the MPLS header on the wire all the way to the egress so QoS EXP bits survive to the last router. Two forwarding tables appear once SR is up: 'mpls.0' is the transit label table, holding the Swap and Pop actions each router applies to labels passing through it, and 'inet.3' is the tunnel table, where each remote loopback resolves to a 'Push <node-SID>' action. BGP next-hops for L3VPN resolve out of inet.3, which is how services get placed on SR transport. Prove the data plane with a labeled 'traceroute mpls segment-routing isis', which walks the actual label-switched path rather than plain IP.

Build steps

1. **Enable SPRING and define the global label block (SRGB).**

```
set protocols isis source-packet-routing srgb start-label 16000 index-range 8000
```

Note: The SRGB is identical on every node: start 16000, range 8000, so the block is 16000 to 23999 domain-wide. This line never changes per node.

2. **Assign this node's unique node-SID index.**

```
set protocols isis source-packet-routing node-segment ipv4-index 1
```

Note: Index 1 is PE1, giving absolute label 16001 (16000 + 1). PE2 uses index 2 (label 16002), P1 index 21, P3 index 23, and so on. Each node's own index lives in its configs/stage2-sr/<node>.set; the index must be globally unique or two nodes claim the same label.

3. **Advertise explicit-null so the label survives to the egress hop.**

```
set protocols isis source-packet-routing explicit-null
```

Note: This requests the explicit-null label (0) on the final hop instead of penultimate-hop popping, keeping the MPLS header, and its EXP bits, on the wire all the way to the destination PE.

4. **Turn on MPLS forwarding and add MPLS to every transit interface plus lo0.**

```
set protocols mpls traffic-engineering mpls-forwarding
set protocols mpls interface lo0.0
set protocols mpls interface ge-0/0/1.0
set protocols mpls interface ge-0/0/2.0
commit and-quit
```

Note: family mpls was already on these interfaces from stage0-base; this stanza is what actually programs mpls.0 and lets the interface switch labels. PE1's transit links are ge-0/0/1 (plane A) and ge-0/0/2 (plane B). Each node lists its own interfaces

in configs/stage2-sr/<node>.set, so repeat per node or push the whole stage with 'python push_lab_configs.py stage2-sr'. After this push, or any later push that touches segment-routing configuration, restart routing on every node in the topology, not only the ones you changed; the SRGB does not take effect until then, and a skipped node leaves IS-IS and BGP looking perfectly healthy while forwarding is silently broken. M7's build walks the full story.

Verification

show route 10.20.255.2 table inet.3 extensive

```
inet.3: 7 destinations, 7 routes (7 active, 0 holddown, 0 hidden)
10.20.255.2/32 (1 entry, 1 announced)
      *L-ISIS Preference: 14
            Next hop: 10.20.0.2 via ge-0/0/1.0, selected
            Label operation: Push 16002
            Label TTL action: prop-ttl
```

What to look for: PE2's loopback now resolves to an SR tunnel in inet.3 with 'Push 16002'. This is the transport L3VPN next-hops resolve over; if inet.3 has no entry for 10.20.255.2, SR has not built the tunnel and no service will forward.

show route table mpls.0 | match 16002

```
16002              *[L-ISIS/14] 00:12:14, metric 30
                  >  to 10.20.0.6 via ge-0/0/1.0, Swap 16002
```

What to look for: Label 16002 (PE2's node-SID) resolves to a single next-hop with a Swap toward PE2. A single authoritative next-hop is correct. If 16002 shows two next-hops at different routers, two nodes are advertising the same index, which is a SID clash.

traceroute mpls segment-routing isis 10.20.255.2

```
Probe options: ttl 30, retries 3
 ttl  Label  Protocol   Address         Previous Hop     Probe Status
   1   16002  SR-ISIS    10.20.0.2       (null)          Success
   2   16002  SR-ISIS    10.20.0.6       10.20.0.2         Success
   3     0   SR-ISIS    10.20.0.10      10.20.0.6        Success
 FEC-Stack-Sent: SR-ISIS
 segment-routing traceroute done
```

What to look for: The labeled traceroute walks the label-switched path hop by hop, each transit router swapping 16002 until the explicit-null (0) label on the final hop, and ends in 'done'. This is the data-plane proof the SR tunnel to PE2 actually forwards, not just that a route exists.

show isis overview | match SPRING|Source

```
SPRING based MPLS: Enabled
Source Packet Routing (SPRING): Enabled
  SRGB Config Range: start-label 16000 index-range 8000
  Node Segments: ipv4-index 1
```

What to look for: SPRING is enabled with the shared SRGB 16000/8000 and PE1's own index 1. Confirm each node reports its own index here; a node showing the wrong or a duplicate index is the root of a label clash.

[screenshot 003: M2 verification: label-switched path PE1 to PE2]

Break / Fix exercise

Scenario: SR is enabled across the lab and IS-IS is fully up, but the transport tunnel to PE2 is misbehaving. From PE1, 'traceroute mpls segment-routing isis 10.20.255.2' either times out or returns an inconsistent hop count that never reaches PE2, even though a plain IP 'ping 10.20.255.2' still succeeds. On a transit router, 'show route table mpls.0 | match 16002' resolves PE2's node-SID label toward a next-hop that is clearly not on the path to PE2, and inet.3 on PE1 shows the 10.20.255.2 tunnel resolving to the wrong router.

Your task: The IP underlay is healthy, so the fault lives in the SR transport layer, not IS-IS. Diagnose why label 16002 is being switched toward the wrong node and restore a clean end-to-end label-switched path from PE1 to PE2, using only source-packet-routing configuration. Confirm the fix with mpls.0 showing a single correct next-hop and a labeled traceroute that completes.

Hint: Adjacency state and IP reachability tell you nothing here; work from the label tables. Label 16002 is PE2's node-SID, and its absolute value is SRGB-start plus an index, so ask which router is advertising that index. Compare each node's advertised node-segment index (visible in 'show isis overview' or the IS-IS database) against the design value for its loopback, and remember that every index in the SRGB must be globally unique.

The instructor holds the fault-injection and fix commands in the answer key. Diagnose it yourself first.

Brief it back

Explain each of these aloud before moving on. If you cannot, re-read the primer and re-run the verification commands.

- Explain how a node-SID label is derived from the SRGB and the node index, and why label 16002 means 'the path to PE2' on every router in the domain.
- What is the difference between mpls.0 and inet.3, and which one does an L3VPN BGP next-hop resolve over to get onto SR transport?
- Why does a labeled 'traceroute mpls segment-routing isis' prove something that a plain 'ping' to the same loopback does not?